GROUP NAME: CODE MONKEYS **SUBJECT:** IT 105 Information Management I

COURSE, YEAR & SECTION: BSIS 2A **PROFESSOR:** Dr. GUILLERMO V. RED, Jr.

NoSQL Reflection Essay

In our Student Management System, the audit_logs table is a good candidate for NoSQL. This table currently stores user actions like INSERT, UPDATE, DELETE, and LOGIN using fixed columns such as logID, action, tableName, and actionTime. However, different actions require different information. For example, a LOGIN action should record IP address and device type, while an UPDATE action should store the old and new values. Using a regular SQL table for this would result in many empty (NULL) columns, which is inefficient.

A document-based NoSQL database like MongoDB is better suited for audit logs for three reasons. First, it offers a flexible schema where each log entry can have its own fields without changing the table structure. Second, MongoDB handles millions of writes per second, making it very fast for logging. Third, logs can be easily spread across multiple servers for scalability.

Here is an example MongoDB document for a login event:

```
"action": "LOGIN",
"user": "user4975",
"role": "staff",
"timestamp: "2026-04-10T10:30:00Z",
"ipAddress": "192.168.1.100",
"device": "Chrome/Windows"
}
```

To find all login logs from the last 7 days, the query would be:

```
db.audit_logs.find({ action: "LOGIN", timestamp: { $gt: "2026-04-03" } })
```

Conclusion

Using MongoDB for audit logs would make our system more flexible and faster. The semi-structured nature of logs fits perfectly with document databases.

Reference:

MongoDB. (2025). *MongoDB is fantastic for logging*. MongoDB Blog. https://www.mongodb.com/company/blog/product-release-announcements/mongodb-is-fantastic-for-logging?91e05470